

is consistent. Column 3 has no pivots. $\therefore$ From Theorem 2, the system of equations has infinitely many solutions.

$\therefore b$ is in the column space of A . $\square$

Definition: The **null space** of a matrix A is the set $\text{Nul } A$ of all solutions of the homogeneous equation $Ax=0$.

Theorem 12: The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.
Equivalently, the set of all solutions of a system $Ax=0$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Proof: By defn, the null space of a $m \times n$ matrix A is the set $\text{Nul } A$ of all solutions of the homogeneous equation $Ax=0$.

We know that $A \cdot 0 = 0$ is the trivial solution of $Ax=0$ where $0 \in \mathbb{R}^n$. $\therefore 0 \in \text{Nul } A$.

Let, $x, y \in \text{Nul } A$. $\therefore Ax=0$ and $Ay=0 \Rightarrow Ax+Ay=0+0$
 $\Rightarrow Ax+Ay=0$ (Algebraic property 3) $\Rightarrow A(x+y)=0$ (Thm 5(a) from Chapter 1)

$\therefore (x+y) \in \text{Nul } A$.

Let, $c \in \mathbb{R}$ and $x \in \text{Nul } A$. $\therefore Ax=0 \Rightarrow c \cdot (Ax) = c \cdot 0 \Rightarrow c \cdot (Ax) = 0$
 $\Rightarrow A \cdot (cx) = 0$ (Thm 5(b) from Chapter 1)

$\therefore cx \in \text{Nul } A$. By Defn, $\text{Nul } A$ is a subspace of $\mathbb{R}^n$. $\square$

Basis for a Subspace

• Because a subspace typically contains an infinite number of vectors, some problems involving a subspace are handled best by working with a small finite set of vectors that span the subspace.

It can be shown that the smallest possible spanning set must be linearly independent.